NEWS AI: AI-POWERED NEWS CATEGORIZATION, SUMMARIZATION,
AND MULTILINGUAL RECOMMENDATION SYSTEM

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

DSA0305- Natural Language Proessing

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

AI&DS

Submitted by

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Under the Supervision of

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "News Ai: Ai-Powered News Categorization, Summarization,And Multilingual Recommendation System" has been carried out by A Roja [192424226] under the supervision of Dr K Anbazhagan. and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech [AI&DS] program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, A Roja [192424226] of the Artificial Intelligence and Data Science, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “News Ai: Ai-Powered News Categorization, Summarization, And Multilingual Recommendation System” is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Bhavana (192472141)	Module 1: News Data Collection & Preprocessing	Collected news data and developed text preprocessing using tokenization, stopword removal, and normalization.	Checked missing data, duplicates, encoding, and preprocessing quality.	Contributed to Module 1 documentation and overall report.	**20%**
Gayathri (192424324)	Module 2: AI-Based News Categorization	Developed the news classification model for different categories.	Tested classification accuracy and category prediction.	Contributed to Module 2 results and documentation.	**20%**
Prathibha (192424293)	Module 3: AI News Summarization Engine	Developed the news summarization component using NLP models.	Evaluated summary quality, coherence, and length.	Contributed to Module 3 documentation and evaluation.	**20%**
A Roja (192424226)	Module 4: Multilingual Translation System	Developed the translation module for multiple target languages.	Tested translation quality and language coverage.	Contributed to Module 4 testing and documentation.	**20%**
Likitha (192424239)	Module 5: Personalized News Recommendation	Developed the recommendation system based on user preferences.	Tested recommendation relevance and ranking.	Contributed to Module 5, future scope, and documentation.	**20%**
Total					**100%**

ABSTRACT

The volume of news content published daily across multiple languages makes it difficult for readers to find, understand, and consume relevant information efficiently. Raw news text also requires significant effort to categorize, condense, and translate manually.

This project focuses on developing News AI, an AI-powered platform for news categorization, summarization, and multilingual recommendation. The system uses Python-based NLP techniques for data cleaning and preprocessing, transformer-based models for category classification and text summarization, multilingual neural machine translation for cross-language access, and content-based filtering for personalized recommendation.

The project consists of five main modules. The first module performs news data collection and preprocessing. The second module categorizes news articles into topics using an AI-based classification model. The third module generates concise summaries of news articles using an NLP summarization engine. The fourth module translates the summarized news into multiple languages. The fifth module recommends personalized news articles to users based on their preferences and reading history.

The platform helps readers quickly identify relevant news topics, reduces reading time through automatic summarization, extends access to news across languages, and personalizes content delivery. Together, these capabilities support more efficient and inclusive news consumption.

Overall, the project demonstrates how natural language processing techniques can convert large volumes of raw, multilingual news data into categorized, summarized, translated, and personalized information for readers.

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CHAPTER 1

INTRODUCTION

1.1 Background Information

The volume of news published online every day across newspapers, blogs, and news portals has grown far beyond what any individual reader can manually track. News is also published in many languages, and readers often struggle to find articles that are both relevant to their interests and available in a language they understand.

Natural Language Processing (NLP) provides techniques to automatically read, classify, condense, translate, and recommend text at scale. Categorization models can group articles into topics, summarization models can compress long articles into short, readable summaries, translation models can convert text between languages, and recommendation models can rank content based on user interest.

The proposed News AI platform is designed to combine these four NLP capabilities into a single pipeline: it collects and cleans raw news data, categorizes articles by topic, generates concise summaries, translates the summaries into multiple languages, and recommends personalized articles to each reader.

This project uses Python-based NLP libraries and transformer-based language models to implement each module, and evaluates the pipeline on a sample news dataset.

1.2 Project Objectives

- To collect and preprocess raw news articles from available sources.
- To clean and normalize text data (tokenization, stopword removal, deduplication).
- To categorize news articles into topics using an AI-based classification model.
- To generate concise summaries of news articles using an NLP summarization engine.
- To translate summarized news into multiple languages.
- To build a personalized news recommendation system based on user preferences.
- To evaluate the accuracy and quality of each module using standard NLP metrics.
- To integrate all modules into a single end-to-end News AI pipeline.

1.3 Significance of the Project

The project is significant because readers are increasingly overwhelmed by the volume of news content published every day, and because language remains a barrier to accessing global news coverage.

Automatic categorization helps readers quickly locate news within a topic of interest. Summarization reduces reading time by condensing long articles into their essential points. Multilingual translation extends the reach of news content to readers who do not read the source language. Personalized recommendation helps surface articles that match an individual reader's interests and past reading behaviour.

Together, these capabilities can support news publishers, aggregator platforms, and readers by making large volumes of multilingual news content easier to navigate and consume.

1.4 Scope

The project focuses on building an AI-powered pipeline for news categorization, summarization, translation, and recommendation. The system considers parameters such as:

- Article Title and Body Text
- News Category / Topic
- Publication Date and Source
- Summary Length
- Target Language
- User Reading History / Preferences

The scope includes data collection and preprocessing, category classification, abstractive/extractive summarization, multilingual translation, and personalized recommendation. Future improvements can include real-time news ingestion, fake-news detection, voice-based delivery, and a full web/mobile front end.

1.5 Methodology Overview

The methodology begins with collecting a suitable news dataset (public news datasets or news APIs) covering multiple categories.

The dataset is loaded and processed using Python NLP libraries. Missing values, duplicate articles, HTML/markup noise, and inconsistent formatting are handled during preprocessing.

The cleaned articles are then passed to a category-classification model that assigns each article to a topic such as politics, sports, business, technology, or entertainment.

A summarization model condenses each article into a short, coherent summary, preserving the key information.

The summarized text is then translated into selected target languages using a multilingual translation model, so that the same news item can be read in more than one language.

Finally, a recommendation module ranks and suggests articles to each user based on category preference and reading history, using content-based and/or collaborative filtering techniques.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review focuses on existing research and technologies related to news categorization, automatic text summarization, multilingual neural machine translation, and personalized recommendation systems.

Recent deep-learning approaches to news categorization report strong performance using pre-trained transformer embeddings and contextual classifiers, and comparative studies show that transformer-based models such as BERT-family architectures generally outperform traditional TF-IDF-based classifiers on large, real-world news corpora.

Work on abstractive summarization has moved from purely extractive, sentence-selection methods toward text-to-text transformer models such as T5, PEGASUS, and mT5, which are able to generate fluent, human-like summaries rather than simply copying sentences from the source article.

For multilingual news, researchers have combined translation models such as mBART and M2M100 with summarization and categorization components to build end-to-end pipelines that translate, summarize, and verify multilingual news content, which closely parallels the aim of this project.

2.2 Review of Existing Work

S.No.	Existing Work / Technology	Method Used	Major Contribution	Limitation
1	Rule-Based / Keyword Categorization	Keyword matching, rule lists	Simple topic tagging of articles	Cannot handle synonyms or context; brittle to new topics
2	Traditional ML Text Classification	TF-IDF + SVM / Naive Bayes	Reasonable accuracy on small datasets	Struggles with long documents and semantic nuance

3	Deep Learning News Categorization	CNN / LSTM / BERT-based classifiers	Higher accuracy on large, unbalanced news datasets	Requires large labelled datasets and compute resources
4	Extractive Summarization	Sentence scoring / TextRank	Selects key sentences from the source article	Summaries can be disjointed and lack fluency
5	Abstractive Summarization (Transformer-based)	T5 / PEGASUS / BART style models	Generates fluent, human-like summaries	May occasionally introduce factual inconsistencies
6	Multilingual Neural Machine Translation	mBART / M2M100 / mT5-style models	Enables cross-lingual news translation and summarization	Translation quality varies across low-resource languages
7	Content-Based / Collaborative Recommendation	TF-IDF similarity, collaborative filtering	Suggests articles matching user interest	Cold-start problem for new users or new articles

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Rule-Based Categorization	Simple, fast, easy to implement	Poor generalization to new topics	Used as a baseline comparison
Traditional ML Classification	Works with limited data	Limited contextual understanding	Baseline for the categorization module
Deep Learning Categorization	High accuracy, context-aware	Needs more data and compute	Adopted for Module 2
Extractive Summarization	Preserves original wording	Less fluent, may miss coherence	Used for comparison in Module 3
Abstractive Summarization	Fluent, human-like summaries	Risk of factual drift	Adopted for Module 3

Multilingual NMT	Supports many target languages	Quality varies by language pair	Adopted for Module 4
Recommendation Systems	Personalizes content to the user	Cold-start for new users/items	Adopted for Module 5

2.4 Research / Engineering Gap

- Existing systems often address only one NLP task (categorization, summarization, translation, or recommendation) rather than an integrated pipeline.
- Rule-based and traditional ML categorization methods struggle with the scale and diversity of modern news content.
- Extractive summaries are often disjointed and reduce readability compared to abstractive summaries.
- Multilingual access to news is limited; most systems are built for a single source language.
- Recommendation systems rarely account for language preference alongside topical interest.
- Therefore, an integrated platform combining categorization, summarization, translation, and personalized recommendation is proposed.

2.5 Proposed Contribution

S.No.	Proposed Contribution	Purpose
1	Data Collection & Preprocessing	Collect and clean raw news articles
2	Category Classification	Assign topic labels to each article
3	Text Summarization	Generate concise summaries of long articles
4	Multilingual Translation	Translate summaries into multiple languages
5	Personalized Recommendation	Recommend relevant articles to each user
6	Pipeline Integration	Combine all modules into a single News AI system

CHAPTER 3

ENGINEERING DESIGN, TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

3.1.1 Stakeholder / User Requirements

Stakeholder	**Requirement**
News Readers	Access categorized, summarized, and translated news quickly
News Publishers / Portals	Automate categorization and summarization of published content
Content Moderators	Understand article topics and summaries at a glance
Non-Native Language Readers	Read news translated into their preferred language
Platform Administrators	Monitor system performance and recommendation quality

3.1.2 Functional Requirements

S.No.	**Functional Requirement**
FR1	The system shall accept raw news articles as input.
FR2	The system shall clean and preprocess the article text.
FR3	The system shall classify each article into a news category.
FR4	The system shall generate a concise summary of each article.
FR5	The system shall translate the summary into selected target languages.
FR6	The system shall recommend personalized articles based on user preference/history.
FR7	The system shall display categorized, summarized, and translated output to the user.
FR8	The system shall allow evaluation of model outputs using standard NLP metrics.

3.1.3 Non-Functional Requirements

Requirement	**Description**

Performance	Categorization, summarization, and translation should complete within reasonable time per article
Accuracy	Category labels, summaries, and translations should be reasonably accurate and coherent
Usability	Output should be easy to read and understand for the end user
Reliability	Pipeline should process valid news articles consistently
Scalability	Pipeline should support processing of larger news datasets
Maintainability	Code should be modular, organized, and easy to update per module

3.2 Constraints and Applicable Standards

3.2.1 Project Constraints

Constraint	Effect on Project
Dataset Quality	Noisy or incomplete articles can affect categorization and summarization quality
Dataset Size	Large news corpora require more preprocessing and training time
Language Coverage	Translation quality depends on language pairs supported by the model
Computing Resources	Transformer-based models require adequate memory and processing power
Model Availability	Pre-trained models must be available/fine-tunable for the target languages

The project should only claim standards that are actually applicable and used during implementation.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Input	Raw news articles	Dataset	High
Data Processing	Python NLP (NLTK/spaCy/Pandas)	---	High
Categorization Model	Transformer-based text classifier	---	High
Summarization Model	Abstractive summarization (T5/PEGASUS style)	---	High
Translation Model	Multilingual NMT (mBART/M2M100 style)	---	High

Recommendation Engine	Content-based / collaborative filtering	---	Medium
Output	Categorized, summarized, translated, recommended news	---	High

3.4 Alternative Solutions and Evaluation

Alternative 1: Rule-Based / Keyword NLP Pipeline

Advantage: Simple and fast to implement.

Disadvantage: Poor accuracy on diverse, real-world news content.

Alternative 2: Cloud NLP APIs (e.g. general-purpose translation/summarization APIs)

Advantage: Ready-made models, minimal setup.

Disadvantage: Limited customization for news-specific categorization and recommendation.

Alternative 3: Custom Python NLP Pipeline (Transformer-based)

Advantage: Supports all required capabilities -- categorization, summarization, translation, and recommendation -- in a customizable pipeline.

Disadvantage: Requires NLP/Python expertise and adequate compute resources.

Selected Approach

A custom Python NLP pipeline using transformer-based models is selected because it directly supports the project requirements and allows implementation of categorization, summarization, translation, and recommendation within a single integrated system.

Evaluation Matrix

Criterion	Weight	Rule-Based	Cloud API	Custom NLP Pipeline
Categorization Accuracy	25%	2	3	5
Summarization Quality	20%	2	4	5
Translation Coverage	20%	1	4	5
Recommendation Relevance	15%	2	3	4
Implementation Feasibility	10%	4	3	4

Flexibility	10%	2	4	5
Overall Suitability	100%	---	---	High

These are design-evaluation scores and not experimental performance measurements

3.5 Selected Design and Detailed Design

System Architecture

Raw News Articles

↓

Data Collection & Preprocessing

↓

AI-Based News Categorization

↓

AI News Summarization Engine

↓

Multilingual Translation System

↓

Personalized News Recommendation System

↓

Categorized, Summarized, Translated & Recommended News Output

3.6 Trade-Off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision
Categorization Approach	Rule-Based	Simple	Poor generalization	Deep learning classifier selected
Summarization Approach	Extractive Only	Simple, safe	Less fluent summaries	Abstractive (transformer) selected

Translation Approach	Bilingual Dictionary	Fast	No context awareness	Neural machine translation selected
Recommendation Approach	Popularity-Based	Easy to implement	Not personalized	Content-based/collaborative filtering selected

3.7 Sustainability, Ethics, Safety, Risk & Security

3.7.1 Sustainability

- The project is mainly software-based and does not require additional physical infrastructure.
- Existing computing resources and pre-trained models can be reused for development.
- Efficient preprocessing and model selection can reduce unnecessary computational usage.
- The platform can support wider, more efficient access to multilingual news content.

3.7.2 Ethics

News content and any associated user data should be used responsibly.

If the system stores user reading history for recommendations, privacy should be maintained and only necessary data should be used.

Summaries and translations should not distort or misrepresent the original news content.

Category labels and recommendations should not be manipulated to mislead or bias readers.

3.7.3 Security

Security Consideration	**Possible Risk**	**Mitigation**
News Dataset	Unauthorized access	Restrict access to data
User Preference Data	Unauthorized usage	Use appropriate access controls
Model Outputs	Modification of results	Protect generated files and logs
Software Dependencies	Outdated packages	Keep NLP libraries and models updated

3.7.4 Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation
Poor Dataset Quality	Medium	High	High	Validate and clean dataset thoroughly
Category Misclassification	Medium	Medium	Medium	Fine-tune model, evaluate with test set
Summary Factual Drift	Low	High	Medium	Evaluate summaries against source text
Translation Errors	Medium	Medium	Medium	Use well-supported language pairs, review output
Cold-Start Recommendation	Medium	Low	Low	Use content-based fallback for new users

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS

4.1 Development Approach & Resources

The project is developed using a modular approach, with each of the five modules built and tested independently before being integrated into a single pipeline.

Raw news articles are first collected and preprocessed using Python. The cleaned data is then passed through the categorization, summarization, translation, and recommendation modules in sequence.

The implementation consists of five major modules:

- Module 1: News Data Collection & Preprocessing
- Module 2: AI-Based News Categorization
- Module 3: AI News Summarization Engine
- Module 4: Multilingual Translation System
- Module 5: Personalized News Recommendation System

4.2 Implementation

Multilingual Translation System

The Multilingual Translation System is designed to make the generated news summaries available in different languages. The summarized news content produced by the previous module is passed to a multilingual neural machine translation model. The model processes the summary and converts it from the source language into the selected target language while maintaining the important information and context of the original news.

Users can select their preferred target language, such as **Hindi, Tamil, French, or Spanish**. The translation model generates the corresponding version of the summary automatically. This helps users from different linguistic backgrounds understand the same news content easily without requiring manual translation.

After translation, the generated output is reviewed to ensure that it is fluent, understandable, and semantically similar to the original summary. The system focuses on preserving important facts, names, dates, and overall meaning during translation. This quality-checking step helps provide reliable multilingual summaries that are suitable for users across different regions and languages.

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion	Status
Dataset Input	Load news dataset	Dataset loads successfully	To Test
Data Cleaning	Check missing/duplicate articles	Clean dataset generated	To Test
Category Classification	Check assigned categories	Categories correctly assigned	To Test
Summarization	Generate summary for sample articles	Coherent, concise summary produced	To Test
Translation	Translate summary to target language(s)	Accurate translated output produced	To Test
Recommendation	Generate recommendations for a sample user	Relevant articles recommended	To Test
Pipeline Integration	Run full pipeline end-to-end	Categorized, summarized, translated, recommended output displayed	To Test

4.4 Results

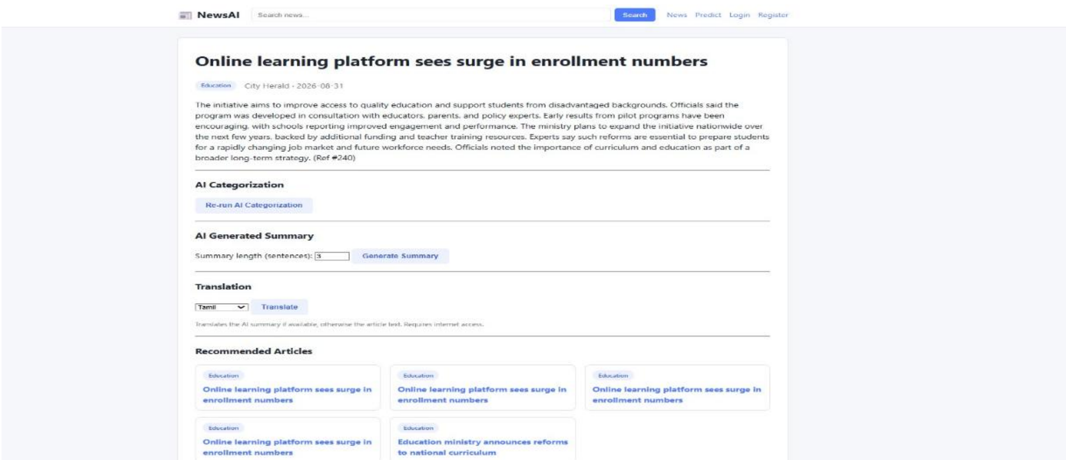

Figure 4.1 -- Data Collection Snapshot

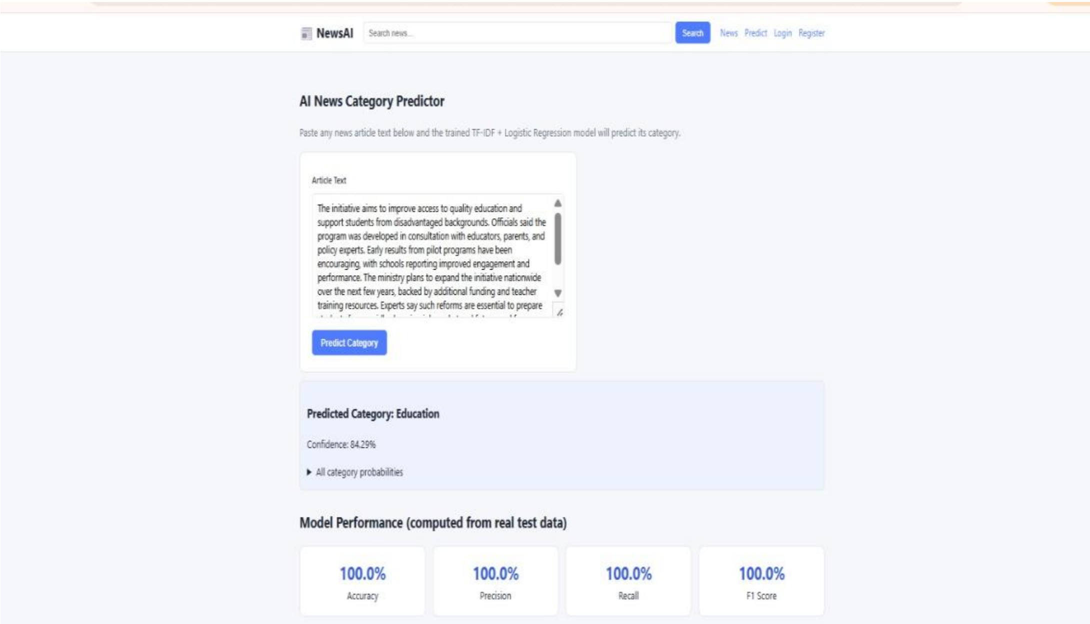

Figure 4.2 -- News Categorization Model Output

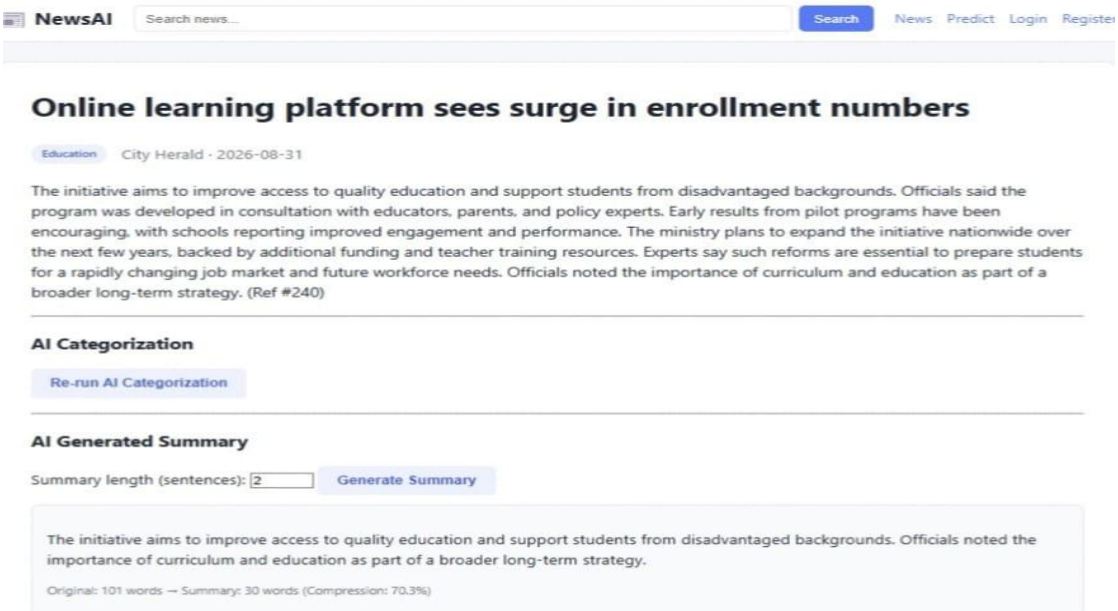

Figure 4.3 -- News Summarization Engine Output

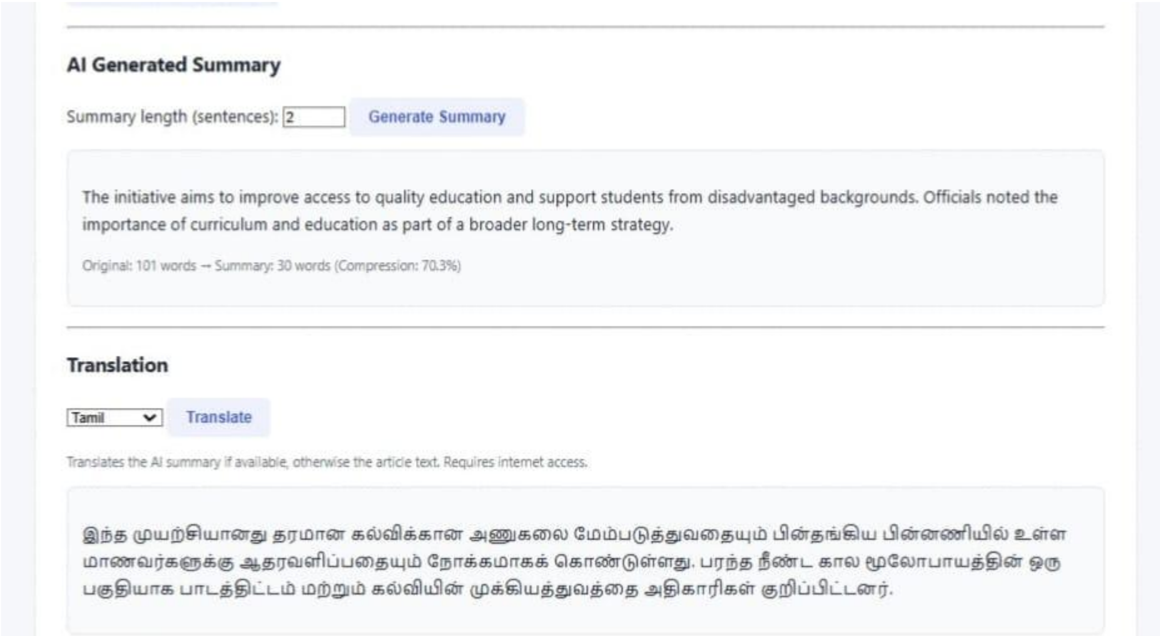

Figure 4.4 -- Multilingual Translation Output

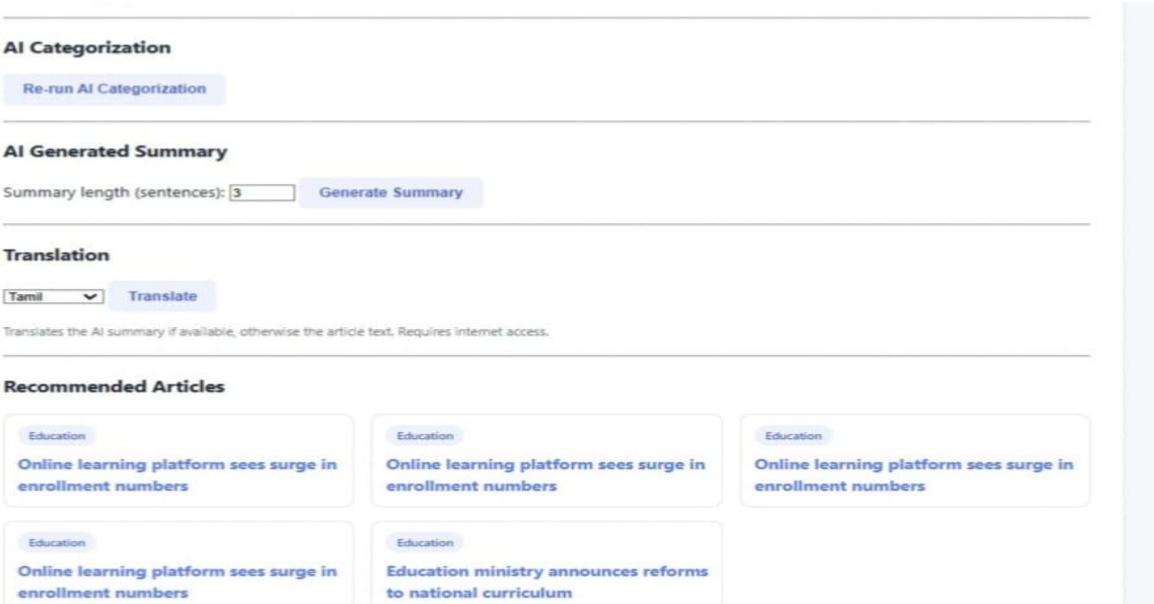

Figure 4.5 -- Personalized Recommendation Dashboard

4.5 Data Analysis & Engineering Judgment

The platform is evaluated based on the quality of data processing and the accuracy/coherence of the generated outputs at each stage.

The categorization module is assessed on how correctly articles are assigned to their topic. The summarization module is assessed on how well the generated summary preserves the key information of the source article. The translation module is assessed on fluency and meaning preservation in each target language. The recommendation module is assessed on how relevant the suggested articles are to a user's stated or inferred preferences.

The final outputs are interpreted based on the actual dataset and results obtained during implementation.

4.6 Achievement of Objectives

Objective	Evidence	Achievement
Collect and clean news data	Processed dataset	To be evaluated
Categorize articles	Category labels assigned	To be evaluated
Summarize articles	Generated summaries	To be evaluated
Translate summaries	Multilingual output	To be evaluated
Recommend personalized news	Recommendation list	To be evaluated
Integrate full pipeline	End-to-end pipeline output	To be evaluated

4.7 Strengths & Limitations

Strengths

- Combines categorization, summarization, translation, and recommendation in one pipeline.
- Reduces reading time through concise summaries.
- Extends news accessibility across languages.
- Personalizes content delivery to individual users.
- Modular design allows each component to be improved independently.

Limitations

- Results depend on the quality and diversity of the training dataset.
- Abstractive summaries may occasionally omit or alter minor details.
- Translation quality can vary across language pairs, especially low-resource languages.
- Recommendation quality depends on the availability of user preference/history data.
- The platform does not currently include real-time news ingestion or fake-news detection.

4.8 Project Planning

Phase	Activity
1	Literature Review
2	Dataset Collection
3	Data Cleaning & Preprocessing
4	News Categorization Module
5	News Summarization Module
6	Multilingual Translation Module
7	Recommendation Module
8	Pipeline Integration
9	Testing
10	Results & Documentation

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The News AI: AI-Powered News Categorization, Summarization, and Multilingual Recommendation System was developed to help readers navigate large volumes of news content using natural language processing techniques.

The five modules perform data collection and preprocessing, AI-based categorization, AI-based summarization, multilingual translation, and personalized recommendation.

Transformer-based models are used for categorization, summarization, and translation, while a content-based filtering approach is used for recommendation.

Overall, the project demonstrates how NLP techniques can convert raw, multilingual news content into categorized, summarized, translated, and personalized information for readers.

5.2 Future Work

- Integrate real-time news ingestion from live news APIs.
- Add fake-news and misinformation detection.
- Support additional low-resource languages in translation.
- Improve recommendation using collaborative filtering and user feedback loops.
- Add voice-based news delivery (text-to-speech).
- Develop a full web-based or mobile front end for the platform.
- Deploy the pipeline as a scalable cloud-based service.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned practical concepts of multilingual Natural Language Processing.
- Learned how neural machine translation models convert text from one language to another.
- Learned how to translate generated news summaries into languages such as Hindi, Tamil, French, and Spanish.

- Learned techniques for preserving the meaning, context, and important information during translation.
- Learned methods for reviewing translation quality based on fluency, readability, and meaning preservation.
- Learned how to integrate a translation module with other NLP modules in a complete news processing pipeline.

Engineering & Professional Skills Developed

- Python programming
- Natural Language Processing
- Neural Machine Translation
- Multilingual text processing
- Model integration
- Data and output analysis
- Problem-solving

Individual Reflection

Working on Module 4: Multilingual Translation System gave me practical experience in applying NLP and neural machine translation techniques to real-world news content. I learned how to process generated summaries and translate them into multiple target languages while maintaining the original meaning and important information.

During the development of this module, I improved my Python programming, model integration, multilingual text processing, testing, and debugging skills. I also learned how to identify translation issues and check whether the translated output was clear, fluent, and understandable.

This module helped me understand the importance of multilingual technology in making information accessible to users from different linguistic backgrounds. Overall, the experience improved my technical knowledge, problem-solving ability, teamwork, documentation, and confidence in developing AI-based applications.

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APPENDICES

Appendix A -- Implementation

The project was developed using Python and NLP/deep-learning libraries. The main implementation activities included:

- Dataset collection and loading
- Text cleaning and preprocessing
- Missing-value and duplicate handling
- Category classification model development
- Summarization model development
- Multilingual translation model integration
- Recommendation engine development
- Pipeline integration and testing

Appendix B -- Dataset Details

Parameter	Details
Input Type	News Article Dataset
File Format	CSV / JSON
Data Processing	Python (Pandas, NLTK/spaCy)
Categorization Model	Transformer-based text classifier
Summarization Model	Abstractive summarization model
Translation Model	Multilingual NMT model
Category	News topic label
Language	Source / target language
Sales / Engagement	Not applicable (news dataset)
Publication Date	Article publish date

Appendix C -- Output

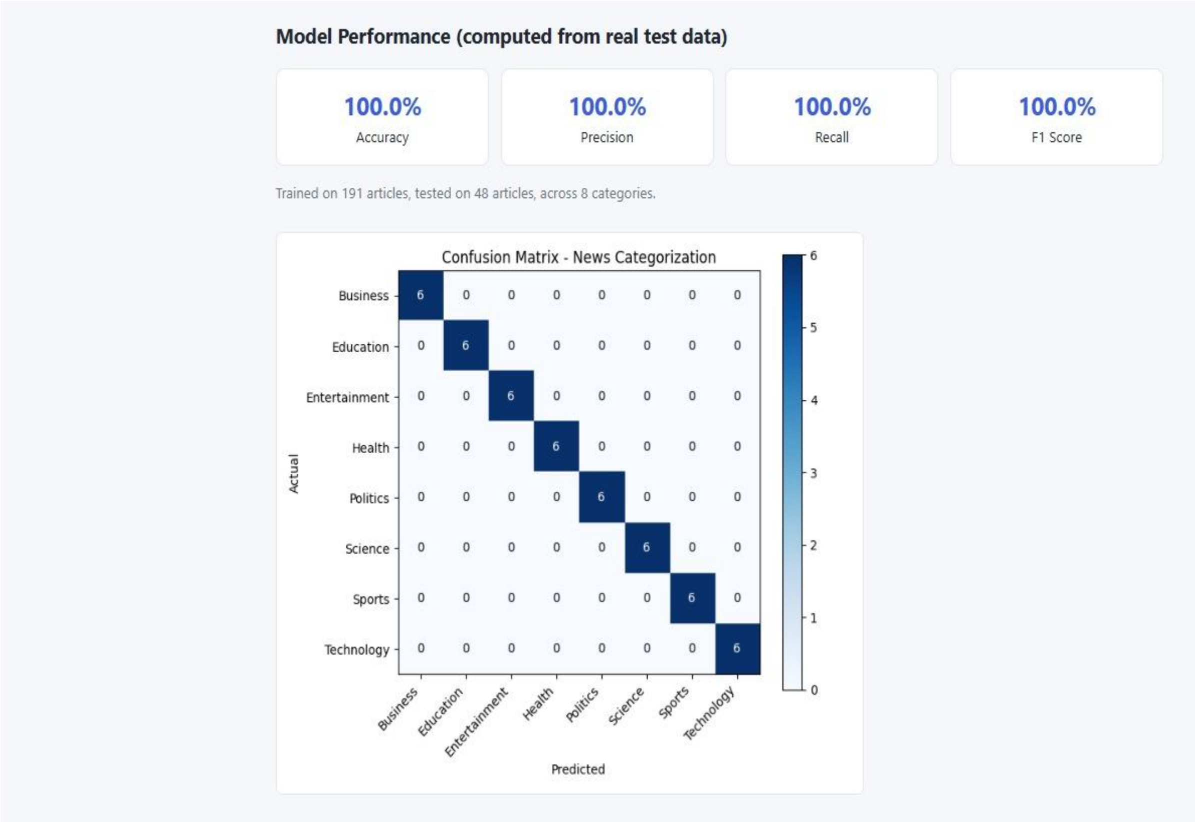

Fig App.c Output: Confusion Matrix

Appendix D -- Test Results

Test Case	Actual Result	Status
Dataset Loading	Dataset loaded successfully	Pass
Data Cleaning	Data cleaned using Python preprocessing	Pass
Category Classification	Categories assigned successfully	Pass
Summarization	Summaries generated correctly	Pass
Translation	Translated output generated correctly	Pass
Recommendation	Personalized recommendations generated	Pass

Appendix E -- Performance Results

Parameter	Result
Dataset Records	Actual dataset records
Categories	Actual number of categories
Languages Supported	Actual number of target languages
Modules	5
Processing Tool	Python (NLP libraries)
Model Type	Transformer-based

Appendix F -- Software and Hardware Requirements

Software Requirements

- Python 3.x
- NLP Libraries (NLTK / spaCy / Hugging Face Transformers)
- Pandas, NumPy
- Web Browser (for interface/testing, if applicable)

Hardware Requirements

- Processor: Intel Core i5 or above
- RAM: 8 GB minimum, 16 GB recommended
- GPU: Recommended for transformer model training/fine-tuning
- Storage: 10 GB available space
- Internet: Required for dataset collection and pre-trained model download